

DSBDL Question Bank – Detailed Answers

1. What is Pandas library in Python?

Pandas is a powerful Python library used for data analysis and data manipulation. It helps in working with tables of data such as Excel sheets or SQL tables. Pandas provides two important data structures: Series and DataFrame. It is widely used in Data Science because it makes handling data easy and fast.

2. List some key features of Pandas.

- Easy handling of missing data
- Fast data filtering and sorting
- Reading data from CSV, Excel, SQL, JSON etc.
- Data cleaning and transformation
- Powerful grouping and aggregation operations
- Time series analysis support

3. What is NumPy Library in Python?

NumPy stands for Numerical Python. It is a library used for mathematical and numerical operations. It provides support for arrays, matrices, and many mathematical functions. NumPy is faster than normal Python lists because arrays are optimized for calculations.

4. What is Matplotlib library?

Matplotlib is a Python library used for data visualization. It helps to create graphs such as line charts, bar charts, histograms, scatter plots, and pie charts. It is useful for understanding data visually.

5. Difference between Seaborn and Matplotlib.

Matplotlib:

- Basic plotting library
- Gives more control over graphs
- Requires more coding

Seaborn:

- Built on top of Matplotlib
- Provides attractive and advanced statistical graphs
- Easier for complex visualization

6. Is Sklearn and Scikit-learn same? What is its use?

Yes, sklearn and Scikit-learn are the same library. sklearn is the short import name. It is used in Data Science for machine learning tasks such as classification, regression, clustering, prediction, and model evaluation.

7. What are functions in Pandas and NumPy?

Common Pandas functions:

- `read_csv()`
- `head()`
- `tail()`
- `describe()`
- `info()`
- `dropna()`

Common NumPy functions:

- `array()`
- `mean()`
- `median()`
- `std()`
- `zeros()`
- `ones()`

8. What is DataFrame in Python?

A DataFrame is a 2-dimensional table-like structure in Pandas. It contains rows and columns similar to an Excel sheet or database table.

9. How to find duplicates in Python?

Pandas command:

`df.duplicated()`

To remove duplicates:

`df.drop_duplicates()`

10. What is use of describe() command?

The `describe()` function gives statistical summary of numerical data such as count, mean, standard deviation, minimum, maximum, and quartiles.

11. Which Naive Bayes classification algorithms are used in Python?

- Gaussian Naive Bayes
- Multinomial Naive Bayes
- Bernoulli Naive Bayes

12. What is significance of Confusion Matrix?

A confusion matrix is used to measure the performance of a classification model. It compares actual values with predicted values and helps to calculate accuracy, precision, recall, and F1-score.

13. What are TP, TN, FP, FN?

- TP (True Positive): Model predicted positive correctly
- TN (True Negative): Model predicted negative correctly
- FP (False Positive): Model predicted positive wrongly
- FN (False Negative): Model predicted negative wrongly

14. What is Recall?

Recall measures how many actual positive values are correctly predicted. Formula: $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

15. What is Precision?

Precision measures how many predicted positive values are actually correct. Formula: $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$

16. What is F1 Score?

F1-score is the harmonic mean of precision and recall. It is useful when data is imbalanced. Formula: $\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

17. What is need of Data Visualization?

Data visualization helps to understand patterns, trends, and relationships in data quickly. It makes data easier to explain and helps in better decision making.

18. What is Outlier?

An outlier is a data value that is very different from other values in the dataset. It may affect analysis and prediction results.

19. When to use Histogram and Pie Chart?

Histogram:
Used to show distribution of continuous numerical data.

Pie Chart:
Used to show percentage or proportion of categories.

20. What are challenges in Big Data Visualization?

- Huge volume of data
- High processing time
- Complex data formats
- Real-time visualization difficulty
- Data quality problems

21. What is Joint Plot and Dist Plot?

Joint Plot:

Shows relationship between two variables.

Dist Plot:

Shows distribution of one variable using histogram and density curve.

22. What are tools used for Data Visualization?

- Matplotlib
- Seaborn
- Tableau
- Power BI
- Excel
- Plotly

23. What is Data Wrangling?

Data wrangling means cleaning, organizing, and transforming raw data into useful format for analysis.

24. What is Data Transformation?

Data transformation means converting data from one format or structure into another suitable format.

25. What is use of StandardScaler function in Python?

StandardScaler is used to standardize data by converting values into a common scale with mean = 0 and standard deviation = 1. It improves performance of machine learning algorithms.

26. What is Hadoop?

Hadoop is an open-source framework used for storing and processing large amounts of data across multiple computers.

27. What is HDFS and MapReduce?

HDFS (Hadoop Distributed File System):

Stores big data across many machines.

MapReduce:

Programming model used to process large datasets in parallel.

28. What are components of Hadoop Ecosystem?

- HDFS
- MapReduce
- YARN
- Hive
- Pig
- HBase
- Sqoop
- Flume

29. What is Scala?

Scala is a programming language that combines object-oriented and functional programming concepts. It runs on JVM (Java Virtual Machine).

30. What are features of Scala?

- Object-oriented language
- Functional programming support
- Runs on JVM
- Less code than Java
- Supports parallel processing

31. How Scala is different from Java?

- Scala supports functional programming; Java mainly object-oriented
- Scala code is shorter
- Scala has better concurrency support
- Scala is more concise and expressive

32. List applications of Scala.

- Big Data processing
- Apache Spark development
- Web applications
- Data analytics
- Machine learning

33. What is Data Science?

Data Science is the field of extracting useful information and insights from data using statistics, programming, and machine learning.

34. What is Big Data?

Big Data refers to extremely large and complex datasets that cannot be handled using traditional data processing methods.

35. What are characteristics of Big Data?

- Volume – huge amount of data
- Velocity – high speed data generation
- Variety – different data formats
- Veracity – uncertainty in data
- Value – useful insights from data

36. List phases in Data Science Life Cycle.

- Data collection
- Data cleaning
- Data exploration
- Model building
- Evaluation
- Deployment

37. What is Central Tendency?

Central tendency refers to the central or average value of a dataset. Mean, median, and mode are measures of central tendency.

38. What is Dispersion?

Dispersion shows how spread out the data values are from the center. Examples are range, variance, and standard deviation.

39. What is Mean, Mode, Midrange, Median? Calculate for given data.

Given data: [10, 22, 13, 10, 21, 43, 77, 21, 10]

Mean = Sum of values / Total values = 25.22

Median = Middle value after sorting = 21

Mode = Most repeated value = 10

Midrange = (Minimum + Maximum) / 2 = 43.5

40. What is Variance?

Variance measures how far data values are spread from the mean. Variance for given data = 431.95

41. What is Standard Deviation?

Standard deviation shows the average spread of data from the mean. Standard deviation for given data = 20.78

42. What is posterior probability in Naive Bayes theorem?

Posterior probability is the probability of an event occurring after considering new evidence. Example: Probability of disease after seeing symptoms.

43. What is likelihood probability in Naive Bayes theorem?

Likelihood probability measures probability of observing data for a given class.

44. How can we deal with missing values?

Methods:

- Remove rows or columns
- Replace with mean, median, or mode
- Forward fill or backward fill
- Use machine learning prediction methods

45. What is NLTK?

NLTK (Natural Language Toolkit) is a Python library used for Natural Language Processing tasks such as tokenization, stemming, and text analysis.

46. What is Tokenization in NLP?

Tokenization means splitting text into smaller parts such as words or sentences.

47. What is Stemming?

Stemming reduces words to their root form. Example: playing → play.

48. What is Lemmatization?

Lemmatization converts words into meaningful base words using dictionary rules. Example: better → good.

49. What is Corpus in NLP?

A corpus is a large collection of text documents used for NLP analysis and training.

50. What is Spark Framework?

Apache Spark is a fast big data processing framework used for large-scale data analytics and machine learning.

51. Steps to run Scala program in Windows with Spark Framework.

1. Copy Scala file into Spark folder.
2. Open CMD in Spark folder.
3. Type spark-shell and press Enter.
4. Load Scala file using command:

```
:load pali.scala  
:load sum.scala
```